

Lisinopril STADA® 20 mg

USE IN PREGNANCY – When used in pregnancy during the second and third trimesters, ACE inhibitors can cause injury and even death to developing fetus. When pregnancy is detected this drug should be discontinued as soon as possible.

Composition

Active ingredient: Each tablet contains 21.78mg Lisinopril dihydrate, equivalent to 20mg Lisinopril.

Excipients: Mannitol, maize starch, calcium hydrogen phosphate dihydrate, pregelatinized starch, colloidal anhydrous silica, magnesium stearate

Pharmacological properties

Lisinopril is an inhibitor of the angiotensin-converting enzyme. The angiotensin converting enzyme (ACE) is a peptidyl dipeptidase, which catalyses the conversion of angiotensin I into the vasoconstricting peptide, angiotensin II. Inhibition of ACE results in a lower concentration of plasma angiotensin II, which causes an increase of the plasma renin activity (caused by the elimination of the negative feedback of angiotensin II on the renin secretion) and reduced aldosterone secretion.

ACE is identical with kininase II. That is why lisinopril can block bradykinin, a strong vasodepressive peptide. Although it is still not clear, which part this plays in the therapeutic effect of lisinopril.

Although the blood pressure reducing mechanism of lisinopril supposedly works primarily by suppression of the renin-angiotensin-aldosterone system, which plays an essential role in the blood pressure regulation, lisinopril has an antihypertensive effect also in patients with a low plasma renin level.

Pharmacokinetics

In clinical trials the maximum plasma concentration was achieved within about 7 hours of oral administration. Patients with an acute myocardial infarction showed a tendency towards a slightly longer time to peak plasma concentration. With repeated doses lisinopril proved to have an effective half-life of 12.6 hours.

The major part of the active agent is eliminated during the early phase, in lower concentrations a prolonged terminal elimination follows, which, however, does not lead to accumulation of the drug. This terminal phase is most likely based on a saturated binding to ACE and is not proportional to the applied dose. It seems that lisinopril will not bind to other serum proteins.

Reduced kidney function reduces the renal elimination of lisinopril. This becomes clinically significant when the glomerular filtration rate is <60ml/min. Elderly patients have higher blood levels and higher AUC values than younger patients. Lisinopril can be removed by dialysis.

Measurements of urine elimination in clinical trials have shown that the average absorbed part of lisinopril is about 29% (25-50%), with an interindividual variation of (6-60%) at all dosages tested (5-80mg).

Lisinopril is not metabolized and the absorbed substance is eliminated completely and unchanged in the urine. The absorption of lisinopril is not affected by the simultaneous intake of food.

Indications

- High blood pressure (arterial hypertension)

- Heart failure (cardiac insufficiency), as an additional treatment when non-potassium sparing diuretics are not sufficient or, if necessary, together with digitalis

- Acute heart attack (myocardial infarction) - of patients with a stable cardiocirculatory situation (haemodynamically stable patients with a systolic blood pressure above 100mmHg, serum-creatinine below 177 µmol/l (2.0mg/dl) and proteinuria less than 500mg/24 hours). Lisinopril should be applied additionally to the usual standard heart attack treatment, preferably together with nitrates.

Side effects

The following undesirable effects were observed during therapy with lisinopril or other ACE inhibitors.

Cardiocirculatory system

Occasionally, especially at the initiation of therapy or when the dosage of lisinopril and/or the diuretics is increased, an excessive drop of blood pressure may occur. This occurs most frequently in patients with salt or fluid deficit after treatment with diuretics, patients with heart failure and severe or renovascular hypertension. Symptoms are dizziness, weakness, impaired vision, and, occasionally, loss of consciousness (syncope).

There are single case reports on the following undesirable effects of ACE inhibitors connected with severe blood pressure fall: tachycardia, palpitations, arrhythmia, chest pain, angina pectoris, heart attack, transitory symptomatic hypoperfusion of the brain (transitory ischemic attack), stroke.

If lisinopril is administered to patients with acute heart attack, especially within the first 24 hours, a second or third degree atrioventricular block and/or a severe hypotension and/or impaired renal function and in rare cases a cardiogenic shock may occur.

Kidneys

Sometimes renal impairment may occur or worsen, in single cases up to the point of an acute renal failure. Rarely proteinuria has been observed, sometimes together with worsening of the renal function.

Respiratory tract

Occasionally dry cough, sore throat, hoarseness and bronchitis may occur, rarely laboured respiration, sinusitis, rhinitis, bronchospasm/asthma, pulmonary infiltration, stomatitis, glossitis and dry mouth were reported. The cough is characteristically non-productive, persistent and reversible after therapy has been discontinued.

In single cases an ACE inhibitor induced angioneurotic oedema involving the larynx, throat and/or tongue lead to a fatal respiratory constriction. There have been single reports of alveolitis (chronic eosinophilic pneumonia) in connection with lisinopril therapy.

Gastro-intestinal tract/liver

Sometimes nausea, abdominal pain and indigestion, rarely vomiting, diarrhea, constipation and lack of appetite occurred.

Rarely during therapy with ACE inhibitors a syndrome was observed that begins with cholestatic jaundice and worsens until hepatic necrosis, which sometimes can be fatal. The mechanism of this syndrome is unknown. Should jaundice develop in patients treated with ACE inhibitors, the therapy is to be discontinued and the patients monitored by a doctor.

Single cases of liver insufficiency, hepatitis, liver failure, pancreatitis and ileus were reported during ACE inhibitor therapy.

Skin, blood vessels

Occasionally allergic skin reactions like rash, rarely urticaria, itching (pruritus) and angioneurotic oedema involving the lips, face and/or the extremities oedema. Single cases of severe skin reactions have been reported including pemphigus, erythema multiforme, exfoliative dermatitis, Stevens-Johnson's syndrome, and toxic epidermal necrolysis (Lyell's-syndrome).

Skin reactions can be accompanied by fever, myalgia, arthralgia, vasculitis and certain changes of laboratory values (eosinophilia, leucocytosis and/or positive ANA).

If a severe skin reaction is suspected a doctor must be consulted immediately and therapy with lisinopril must be discontinued.

Single cases of skin reactions resembling psoriasis, photosensitivity, flush, perspiration, hair loss, nail loss (onycholysis) and worsening of the vascular cramps in Raynaud-syndrome were observed during ACE inhibitor therapy.

Nervous system

Occasionally headache and tiredness may occur, rarely drowsiness, depression, sleep disorders, impotence, tingle, peripheral neuropathy involving paraesthesias, disturbance of balance, muscle cramps, nervousness, confusion, tinnitus, blurred vision and taste disorders (parageusia) or temporary loss of the sense taste (ageusia) were reported.

Findings in laboratory tests (blood, urine)

Occasionally drops in haemoglobin concentration, haematocrit, leucocytes' or thrombocytes' count have been reported. Rarely patients, especially those with reduced kidney function, collagen disease or concurrent therapy with allopurinol, procainamide or certain immunosuppressive medicines, experienced pathological or other changes in the blood count, e.g. anemia, thrombocytopenia, neutropenia, eosinophilia, and in single cases agranulocytosis or pancytopenia.

There have been single reports of patients suffering from congenial G6PD-deficiency experiencing haemolytic anemia.

Rarely, especially in patients with renal impairment, severe heart failure and renovascular hypertension the serum concentrations of urea, creatinine and potassium can increase and the sodium concentration can decrease. In patients with diabetes mellitus hyperkalemia may occur.

In special cases the proteinuria in the urine can be increased.

Single cases of increased serum liver enzymes and bilirubin concentrations have been reported.

Precautions:

The above-mentioned laboratory parameters should be controlled regularly during therapy with Lisinopril STADA® 20mg.

Serum electrolytes and creatinine and the complete blood count should be monitored at short intervals especially in the initial phase of therapy and in high-risk patients (renal impairment, collagen disease) and in those receiving concurrent treatment with immunosuppressants, cytotoxic antineoplastic drugs, allopurinol or procainamide.

Patients experiencing such symptoms as fever, lymph node swelling and/or sore throat in the course of Lisinopril STADA® 20mg therapy should have a white blood cell count without delay.

The doctor or pharmacist should be informed if any adverse effect not described in this leaflet is experienced.

Countermeasures in the event of an adverse experience

If a tissue swelling of the face, the extremities, the lips, the tongue, the glottis and/or the larynx occurs, treatment with Lisinopril STADA® 20mg must be discontinued immediately and a suitable treatment of the patient must be initiated.

If a severe skin reaction is suspected the doctor must be consulted immediately.

Special warnings and precautions for use

It is recommended to initiate lisinopril-therapy under hospital conditions for patients:

- Undergoing a combined and highly dosed therapy with diuretics (e.g. more than 80mg furosemide)
- With fluid or salt deficit (hypovolemia or hyponatremia: serum sodium less than 130mmol/l)
- With pre-existing hypotension
- With instable cardiac insufficiency
- With reduced kidney function
- Undergoing highly dosed therapy with vasodilators
- 70 years or older

Especially at the initiation of therapy and in high-risk patients (patients with renal insufficiency; with collagen disease) and when immunosuppressives, cytostatics, allopurinol and procainamide are used concomitantly the serum creatinine concentrations and the blood count should be checked.

Dual blockade of the renin-angiotensin-aldosterone system (RAAS)

There is evidence that the concomitant use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren increases the risk of hypotension, hyperkalemia and decreased renal function (including acute renal failure). Dual blockade of RAAS through the combined use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren is therefore not recommended (see Interactions).

If dual blockade therapy is considered absolutely necessary, this should only occur under specialist supervision and subject to frequent close monitoring of renal function, electrolytes and blood pressure.

ACE-inhibitors and angiotensin II receptor blockers should not be used concomitantly in patients with diabetic nephropathy.

Hypotension

Lisinopril STADA® 20mg may, especially after the first administration, cause a severe drop of blood pressure. A symptomatic hypotension is rarely seen in patients with uncomplicated hypertension. It is more often seen in patients with an electrolyte or fluid deficit on treatment with diuretics, on a low salt diet, after vomiting or diarrhea or after dialysis and was reported mainly in patients with severe heart failure, with or without resulting renal insufficiency as well as in patients, undergoing treatment with high doses of loop diuretics and patients with sodium depletion (hyponatremia) or reduced kidney function. Therapy with Lisinopril STADA® 20mg for these patients must be initiated under close medical supervision, preferably under hospital conditions with low doses and careful dosage adjustment. Simultaneously kidney function and the serum potassium level must be monitored. If possible, the treatment with diuretics should be discontinued.

This also applies to patients with angina pectoris or cerebrovascular angiopathy, for which an excessive drop of blood pressure may lead to a heart attack or to a stroke.

If hypotension occurs the patient should be placed in a lying position and possibly given liquids orally or intravenously (volume substitution). For the treatment of associated bradycardia the administration of atropine may be necessary. After a successful treatment of the initial dose caused hypotension, there is no need to abandon a further careful dose adjustment with this medication. Should a non-acute hypotension in patients with heart failure become symptomatic it may become necessary to reduce the dosage and/or discontinue the treatment with the diuretic and/or lisinopril. If possible, the treatment with diuretics should be discontinued 2-3 days prior to the start of therapy with lisinopril.

Hypotension with acute myocardial infarction

Treatment with lisinopril must not be started in patients with acute myocardial infarction, if, because of a prior treatment with a vasodilator, there is a risk of further serious haemodynamic deterioration. This applies to patients with a systolic blood pressure of 100mmHg or less or with cardiogenic shock. The maintenance dose should be reduced to 5mg or temporarily to 2.5mg, if the systolic blood pressure is 100mmHg or lower. Therapy with lisinopril can lead to a severe hypotension in patients with acute myocardial infarction. If the hypotension persists (systolic blood pressure below 90mmHg for more than 1 hour) treatment with lisinopril must be discontinued.

Patients with severe cardiac insufficiency after an acute myocardial infarction should receive lisinopril only, if these patients are haemodynamically stable.

Renovascular hypertension/renal artery stenosis

For patients with renovascular hypertension and existing renal artery stenosis, on either side or one side (single kidney) there is a higher risk for a severe hypotension and renal insufficiency through the use of lisinopril. A therapy with diuretics can add to the risk. Renal failure can be accompanied by only a slight change of the serum creatinine values, even in patients with one sided renal artery stenosis. Therapy of such patients must, therefore, be medically closely monitored in a hospital and initiated with a low dose and dosage carefully increased stepwise. Therapy with diuretics should be discontinued and the kidney function closely monitored during the first week of therapy.

Renal insufficiency

Patients with severe renal insufficiency (creatinine clearance <30ml/min) and dialysis patients must not use lisinopril. Lisinopril should be administered carefully to patients with reduced renal function. For these patients a reduced dose or a prolonged dosing interval may be necessary.

A connection between lisinopril therapy and renal failure has been reported mainly in patients with severely reduced cardiac performance or existing renal dysfunction (including renal artery stenosis). If diagnosed in time and treated correctly, renal failure through lisinopril treatment is usually reversible. In some hypertensive patients that had no apparent renal impairment, increases in blood urea and creatinine levels were observed when treated with lisinopril and diuretics at the same time. A dose reduction of the ACE inhibitor or discontinuation of the diuretic may be necessary and an undiagnosed renal arterial stenosis should be considered.

Acute myocardial infarction treatment with lisinopril must not be initiated in patients with signs of renal dysfunction, defined as a serum creatinine concentration over 117 µmol/l (2.0 mg/l) or above and/or proteinuria with more than 500 mg per day. If renal dysfunction develops during treatment with Lisinopril STADA® 20mg (serum creatinine clearance <30 ml/min, or a doubling of the serum creatinine level measured before treatment) lisinopril must be discontinued.

Haemodialysis

For patients on continued haemodialysis lisinopril is contraindicated.

In concomitant use of lisinopril and poly(acrylonitrile, sodium-2-methylallylsulfonate)-high-flux-membrane (e.g. AN 69) in dialysis or haemofiltration there is a risk of a hypersensitivity reaction (anaphylactoid reaction) up to shock. First signs of this anaphylaxis are facial swelling, flush, hypotension and dyspnea. The symptoms usually appear within a few minutes after starting haemodialysis. It is advisable, therefore, to use a different dialysis membrane or a different medicine for the treatment of the high blood pressure or reduced cardiac performance.

Increased serum potassium level (hyperkalemia)

Especially for patients with pre-existing reduced kidney or heart function treatment with lisinopril could lead to an increased potassium level (hyperkalemia). Treatment with potassium sparing diuretics or potassium preparations as supplementary therapy is not advisable, since this could lead to a significant increase of the serum potassium level. However, should concomitant treatment with the above mentioned preparation seem indicated, the serum potassium level should be monitored regularly during therapy.

Elderly patients

The response to ACE inhibitors of elderly patients is possibly better than that of younger patients. Elderly patients should therefore, be treated cautiously. For patients 65 years or older an initial dose of daily 2.5 mg lisinopril as well as monitoring of blood pressure a kidney functions is advised.

Children

The efficacy and safety of lisinopril in children has not been sufficiently documented, therefore, the treatment of children is not advised.

Primary hyperaldosteronism

Patients suffering from primary hyperaldosteronism usually do not respond to antihypertensives, which efficacy is based on the inhibitions of the renin-angiotensin-system. Therefore, the use of lisinopril is not advisable.

Proteinuria

There have been rare cases of patients, especially with reduced kidney functions or after the administration of fairly high doses of lisinopril, developing proteinuria. With clinically relevant proteinuria (more than 1 g/day) Lisinopril STADA® 20mg should be used only after critically weighing expected benefits against potential risks and with regular monitoring of the clinical and laboratory parameters.

LDL lipid apheresis/Desensitisation

During a LDL (low-density lipoprotein) apheresis with dextran sulfate, concomitant use of an ACE inhibitors can lead to life-threatening anaphylactic reactions. Life-threatening anaphylactic reactions (e.g. drop of blood pressure, dyspnea, vomiting, allergic skin reactions) can also appear if lisinopril is administered during a desensitisation therapy against insect stings (e.g. bees/wasps).

Should a LDL apheresis or a desensitisation therapy against insect stings become necessary, lisinopril should be temporarily replaced by a different medicine (but no ACE inhibitors) against hypertension or myocardial insufficiency.

Tissue swelling/angioneurotic oedema

There have been rare reports of tissue swelling of the face, the extremities, the lips, the tongue, the glottis and/or the larynx in patients treated with ACE inhibitors, including lisinopril. They can appear at any time during therapy. In such cases treatment with Lisinopril STADA® 20mg must be discontinued immediately, and suitable monitoring of the patients should be instituted.

In cases where the swelling is limited to the face and lips the condition usually clears up without treatment, although antihistamines are beneficial in relieving the symptoms.

Patients with an anamnestic known angioneurotic oedema not associated with ACE inhibitor treatment could be at a higher risk of developing an angioneurotic oedema during ACE inhibitor treatment.

Angioneurotic oedema involving the tongue, glottis or larynx can be life threatening. Emergency treatment including an immediate subcutaneous administration of 0.3-0.5 mg epinephrine or a slow intravenous administration of 0.1 mg epinephrine (please follow dilution instruction) should be initiated with ECG and blood pressure monitoring. The patients must be hospitalized. A suitable monitoring for at least 12 to 24 hours must follow, to ensure that all symptoms have regressed completely before releasing the patient.

Aortic stenosis/hypertrophic cardiomyopathy

ACE inhibitors should be used with caution in patients with an obstruction in the left -ventricular emission tract. Should the obstruction be haemodynamically relevant lisinopril is contraindicated.

Neutropenia/agranulocytosis

Rare cases of hypertensive patients with neutropenia or agranulocytosis were observed during treatment with ACE inhibitors. This occurred rarely in patients with an uncomplicated course of hypertension, but more often in patients with impaired renal function, especially if they also suffered from a disorder of the vascular or connective tissue system (eg. systemic lupus erythematosus or dermatosclerosis) or under concurrent immunosuppressive therapy. For these patients the white blood count must be controlled regularly. Neutropenia and agranulocytosis are reversible after discontinuation of the ACE inhibitor.

Should fever, swelling of the lymphatic glands and /or sore throat occur during therapy with Lisinopril STADA® 20mg, the white blood count should be checked immediately.

Operation/anaesthesia

Lisinopril blocks the formation of angiotensin II as a consequence of the compensating secretion of renin in patients, who undergo a major operation or are anaesthetised with substances that lower the blood pressure. The resulting increased lowering of the blood pressure can be corrected by volume expansion.

Use during pregnancy or breastfeeding

The use of Lisinopril STADA® 20mg during pregnancy is contraindicated.

It is to be made certain that women of a childbearing age are not pregnant before treatment with Lisinopril STADA® 20mg. During therapy with Lisinopril STADA® 20mg these women must take suitable contraceptive measures.

For women who still become pregnant during therapy with Lisinopril STADA® 20mg, a different, less dangerous for the child, therapy must be administered under medical advice, because the use of Lisinopril STADA® 20mg, especially during the last 6 months of pregnancy, could harm the infant.

ACE inhibitors can be secreted into breast milk, and the effect on the nursing baby has not been examined. Therefore, a baby should not be nursed during therapy with Lisinopril STADA® 20mg.

Effects on ability to drive and use machines

Studies on the effect of this medication on the ability to drive a car are not available. It should, however, be taken into consideration that the ability to drive a car, to use machines or to work without a firm support can be effected through the sometimes occurring dizziness and fatigue.

Interactions with other drugs and other forms of interaction

There have been reports on the following interactions on concurrent use of Lisinopril STADA® 20mg and:

- Diuretics:

Normally the concurrent administration of a diuretic to treatment with lisinopril will additionally enhance the antihypertensive effect of lisinopril. Patients on treatment with diuretic, especially those patients with recently initiated diuretic treatment, can sometimes experience a fall in blood pressure when lisinopril-therapy is added. The risk of symptomatic hypotension during treatment with lisinopril can be reduced by the discontinuation of the diuretic before the start of treatment with lisinopril.

- Potassium sparing diuretics or administration of potassium preparations as supplementary therapy:

After administration of potassium sparing diuretics the potassium levels can experience a further increase, especially in patients with reduced kidney function. ACE inhibitors decrease the potassium elimination caused by the diuretics. Potassium sparing diuretics, e.g. spironolactone, triamterene or amiloride, potassium preparations or potassium containing salt supplements can increase the serum potassium concentration significantly. Should an existing potassium deficiency make the administration of such preparations seem necessary, they should be administered with caution and close monitoring of the serum potassium levels.

- Sodium chloride (salt): Lowers the hypotensive and the heart failure relieving effect of lisinopril.

- Antihypertensives: Enhance the blood pressure reducing effect of lisinopril, especially diuretics.

- Painkillers/drugs for inflammatory conditions (non-steroidal anti-inflammatory drugs e.g. aspirin, indomethacin): Possibly lower the blood pressure reducing effect of lisinopril.

- Lithium: Therapy with this and other medicines that increase the elimination of sodium, the elimination of lithium can be reduced. The serum lithium concentration should, therefore, be monitored carefully, if lithium salts are administered.

- Alcohol: ACE inhibitors enhance the effect of alcohol. Alcohol increases the blood pressure reducing effect of ACE inhibitors.

- Anaesthetics, narcotics, hypnotics (sleeping pills): Increase the blood pressure fall (the anaesthesiologist must be informed about the lisinopril therapy)

- Sympathomimetics can lower the blood pressure reducing effect of ACE inhibitors.

- It has been determined that concurrent use of allopurinol, medication that suppresses the defense reaction (cytostatics, immunosuppressives, systemic corticoids) and procainamide increase the risk for a leucopenia.

- Oral antidiabetics (e.g. sulfonyl urea, biguanides) and insulin: ACE inhibitors can enhance the blood sugar reducing effect of the antidiabetics, especially in the first week of the concurrent administration.

- Antacids can reduce the bioavailability of ACE inhibitors.

- Clinical trial data has shown that dual blockade of the renin-angiotensin-aldosterone-system (RAAS) through the combined use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren is associated with a higher frequency of adverse events such as hypotension, hyperkalemia and decreased renal function (including acute renal failure) compared to the use of a single RAAS-acting agent (see Contraindications and Warnings/Precautions).

Please bear in mind that these precautions may also apply to recently used medicines.

Contraindications

- With renal artery stenosis (bilateral or unilateral in patients with a solitary kidney)

- With hypersensitivity to lisinopril or other ACE inhibitors or one of the other ingredients

- With a tendency for facial swelling (hereditary/idiopathic angioneurotic oedema and anamnestically known angioneurotic oedema as a result of an earlier treatment with ACE inhibitors.

- In condition after kidney transplantation

- With severe kidney dysfunction (kidney insufficiency: creatinine clearances less than 30ml/min)

- During haemodialysis

- With aortic or mitral stenosis or other interruption of outflow from the left heart chamber, e.g. hypertrophic cardiomyopathy, if the blood flow is significantly impaired

- In patients with an instable cardiocirculatory situation after an acute heart attack (haemodynamically instable patients after acute myocardial infarction)

- With systolic blood pressure of 100mmHg or below before therapy with lisinopril

- With cardiogenic shock

- During lactation

- Pregnancy

- Concomitant use of lisinopril and poly(acrylonitrile, sodium-2-methylallylsulfonate)-high-flux-membrane (e.g. AN 69) in an emergency dialysis, because of the risk of a life-threatening hypersensitivity reaction (anaphylactoid reaction) up to shock. This combination, therefore, must be avoided by using different medicines (but no ACE inhibitors) for the treatment of the hypertension or myocardial insufficiency or through the use of a different dialysis membrane.

- The concomitant use of lisinopril with aliskiren-containing products is contraindicated in patients with diabetes mellitus or renal impairment (GFR < 60 ml/min/1.73m²) (see Interactions).

Dosage and administration

After the first application of Lisinopril STADA® 20mg, an excessive hypotension may occur in high-risk patients (patients with a salt and/or fluid deficit e.g. after dialysis, vomiting, diarrhea, on concomitant diuretics-therapy, for patients with heart failure or severe or renovascular hypertension).

Therefore, if possible, the salt and/or fluid deficit should be compensated before the start of treatment with Lisinopril STADA® 20mg and/or the treatment with diuretics discontinued or reduced 2-3 days prior to the start of therapy with the ACE inhibitor. For these patients therapy should begin with the smallest single dose of 2.5mg lisinopril once daily in the morning. Lower strength dosage forms (2.5, 5 or 10mg lisinopril) are available for such patients.

These patients must be medically monitored for at least 8 hours - preferably in a hospital - after the first dose is applied, but also after every increase of dosage of lisinopril and/or the diuretic to avoid an uncontrolled occurrence of an excessive drop of blood pressure. This also applies to patients with angina pectoris or cerebrovascular occlusive disease (cerebrovascular angiopathy), for which an excessive drop of blood pressure may lead to a myocardial infarction or to a cerebrovascular accident.

For patients with malignant hypertension or severe myocardial insufficiency the initiation of therapy with Lisinopril STADA® 20mg should take place in the hospital.

Unless prescribed otherwise, the following dosage regimen is recommended:

Arterial hypertension

Therapy should be initiated with ¼ tablet Lisinopril STADA® 20mg once daily in the morning (equivalent of 5mg lisinopril/day). The dose should be adjusted until the optimal blood pressure is reached. Dosage should not be increased before 3 weeks have elapsed. The usual maintenance dose is ½-1 tablet Lisinopril STADA® 20mg (equivalent of 10-20mg lisinopril) once daily. However, doses of 40mg lisinopril once daily may be administered.

A low initial dose of 2.5mg lisinopril once daily in the morning is necessary for patients with impaired renal function, heart failure, for patients who cannot tolerate a discontinuation of diuretic therapy, for patients with hypovolemia and/or salt deficit (eg. after vomiting, diarrhea or diuretic therapy), or for patients with severe or renovascular hypertension and elderly patients. Lower strength dosage forms (2.5, 5 or 10mg lisinopril) are available for such patients.

Heart failure

Lisinopril can be administered concomitantly with diuretics and digitalis.

The initial dose is 2.5mg lisinopril once daily in the morning. The maintenance dose should be adjusted through a stepwise increase of 2.5mg lisinopril each. Lower strength dosage forms (2.5, 5 or 10mg lisinopril) are available for such patients. An increase of the dosage must be done slowly according to the individual response of the patient.

The time lapse between dosage increases should be at least two, preferably four weeks. The maximum dose of 35mg lisinopril per day should not be exceeded.

Acute myocardial infarction - in haemodynamically stable patients

Lisinopril should be administered additionally to nitrates (eg. intravenously, percutaneously as nitrate adhesive patch) and additionally to the usual standard therapy of infarct. Treatment with lisinopril can be started within 24 hours after the appearance of the first symptoms as long as the patients are haemodynamically stable.

The initial dose is ¼ tablet Lisinopril STADA® 20mg (equivalent of 5mg lisinopril), followed by ¼ tablet (5mg lisinopril) after 24 hours, and ½ tablet (10mg lisinopril) after 48 hours. After that, the dose is ½ tablet Lisinopril STADA® 20mg (equivalent of 10mg lisinopril) daily. Patients with a low systolic blood pressure (<120 mmHg) at the start of treatment or during the first 3 days after the infarct should be given a lower dose - 2.5mg lisinopril. Lower strength dosage forms (2.5, 5 or 10mg lisinopril) are available for such patients.

In the case of hypotension (systolic blood pressure lower than 100mmHg) the daily maintenance dose of ¼ tablet Lisinopril STADA® 20mg (equivalent of 5mg lisinopril/day) should not be exceeded and can, if necessary, be reduced to 2.5mg lisinopril daily. If the hypotension (systolic blood pressure lower than 90 mmHg for more than 1 hour) persists, although the daily dose was reduced to 2.5mg, lisinopril must be discontinued.

The maintenance therapy should be continued for six weeks. The lowest maintenance dose is ¼ tablet Lisinopril STADA® 20mg daily (equivalent of 5mg lisinopril/day). Treatment with lisinopril should be continued in patients with symptoms of heart failure.

Lisinopril STADA® 20mg is compatible with concurrent intravenous or percutaneous (nitrate adhesive patch) treatment with nitroglycerine (glycerine trinitrate).

Dosage for moderately reduced renal function (creatinine clearance 30-70 ml/min) and elderly patients (over 65 years)

The initial dose is 2.5mg lisinopril once daily in the morning, the maintenance dose, dependent upon the blood pressure control, is usually ¼-½ tablet Lisinopril STADA® 20mg daily (equivalent of 5-10mg lisinopril/day). The maximum dose of 1 tablet Lisinopril STADA® 20mg (equivalent of 20mg lisinopril) per day should not be exceeded.

Method of administration and duration of therapy

Lisinopril STADA® 20mg can be taken independently of meals but should be taken with a sufficient amount of water. The described daily dose should as a rule be taken once a day in the morning.

For an easier individual dosing Lisinopril STADA® 20mg is available in the form of a snap tab. The Lisinopril STADA® 20mg snap tab is placed on a hard surface with the centre groove facing upward. The snap tab will break into four equal pieces through exerting light pressure from the top with the thumb.

If a dose is missed or less than prescribed dose is taken, Lisinopril STADA® 20mg therapy should be continued as prescribed. Do not double the dose the next time. If treatment is interrupted or stopped early, patients with high blood pressure may experience a rebound blood pressure increase, and those with heart failure may suffer recurrence of symptoms.

The duration of therapy will be decided by the treating doctor.

Symptoms and management of overdosage

Data on overdosing in humans is not available. Depending on the extent of the overdose the following symptoms are likely: severe hypotension, shock, bradycardia, electrolyte disorders and renal failure.

The usual treatment is an infusion of physiological sodium chloride solution. Lisinopril, the active ingredient of Lisinopril STADA® 20mg, can be removed from the blood by haemodialysis. After an overdose the patient should be watched very closely, preferably in an intensive care setting. The serum electrolytes and creatinine are to be checked frequently. If the overdose occurred only recently, absorption preventing measures should be initiated, for instance gastric lavage, administration of medication that binds the active ingredient (absorbents) and sodium sulfate within 30 minutes of ingestion as well as measures that promote elimination.

If hypotension occurs the patient should be placed in a lying position and an intravenous sodium and volume substitution should be given immediately. Treatment with angiotensin II can be considered. If a bradycardia develops atropine should be administered, and possibly a pacemaker therapy should be performed. ACE inhibitors can be removed from the blood by haemodialysis. The use of high flux polyacrylonitrile dialysis membranes should be avoided.

Shelf life and storage instructions

The expiry date of this pack is printed on the box. Do not store above 30°C.

Keep all medicines out of the reach of children.

Preparations and pack sizes

Lisinopril STADA® 5 mg appears as white, round, biconvex tablet with imprint "5" on one side and scored on both sides, blister-packed in PVC/PVDC/Alu foil blister and packed in cardbox (packs of 30, 50 and 100 tablets).

Lisinopril STADA® 10 mg appears as white, round, biconvex tablet with imprint "10" on one side and quadrisectioned on both sides, blister-packed in PVC/PVDC/Alu foil blister and packed in cardbox (packs of 30, 50 and 100 tablets).

Lisinopril STADA® 20 mg appears as white, round, biconvex tablet with imprint "20" on one side and quadrisectioned on both sides, blister-packed in PVC/PVDC/Alu foil blister and packed in cardbox (packs of 30, 50 and 100 tablets).

Not all strengths and pack sizes are available in certain countries.

Manufactured under the license of:

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